Introduction → Some advanced ML (↙ Prod.) Algo

Expectations → MWF → (9pm - 11:15 pm)

→ WA group

→ Live classes + PSP * (Top 20 percentile)

→ Industry experts

- Rohit Jindal (Lead Data Scientist)

→ D.C.E / IISc Bangalore

→ Mastercard AI Garage, SSGA,

→ Target (US, MN)

12 sessions → KNN - 1

Agenda → Case study (P.S) (Blinkit Case Study)

→ Quiz (Introduce K Nearest Neighbors)

Geometric

Scratch Implementation (>50LPA)

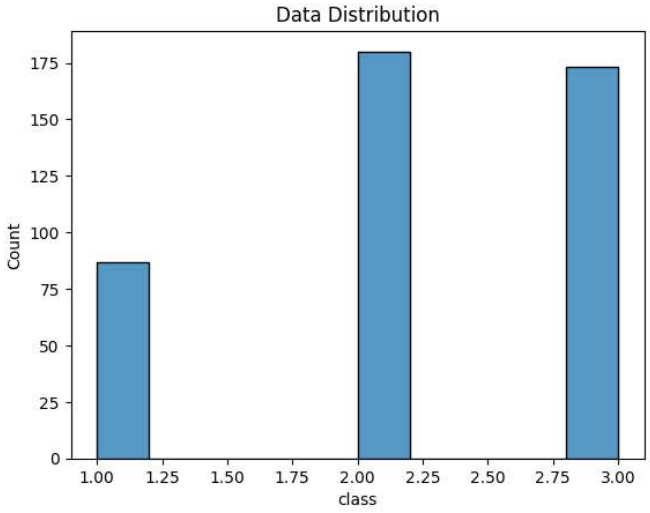
BUSINESS CASE - BLINK IT

Blink it needs optimal number of delivery partners for each store.

Hence, classified stores into 3 based on outgoing deliveries.

- High Traffic — C1
- Moderate Traffic — C2
- Low Traffic — C3





Q: which clf algo do you know ?

② [Log. Reg] → Linear decision Boundary

$$\log(\text{odds}) = w^T x$$

$$\log\left(\frac{p}{1-p}\right) = w^T x$$

(Binary) ↓

(Simply Linear Case)

multi class

(OVR)

(PCA does)

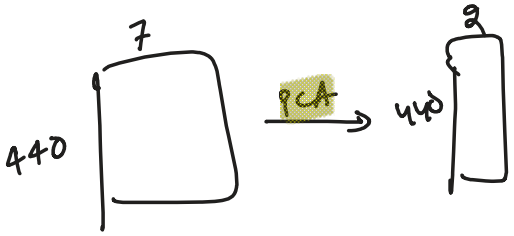
Inputs 7 (7d) → (2d)

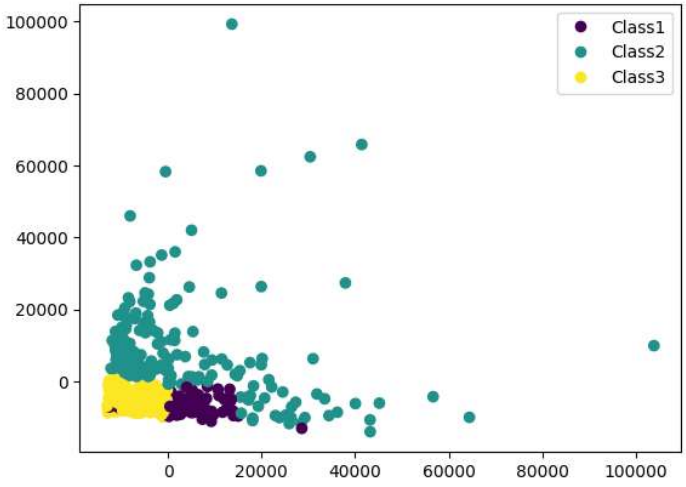
Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
3	12669	9656	7561	214	2674	1338
3	7057	9810	9568	1762	3293	1776
3	6353	8808	7684	2405	3516	7844
3	13265	1196	4221	6404	507	1788
3	22615	5410	7198	3915	1777	5185

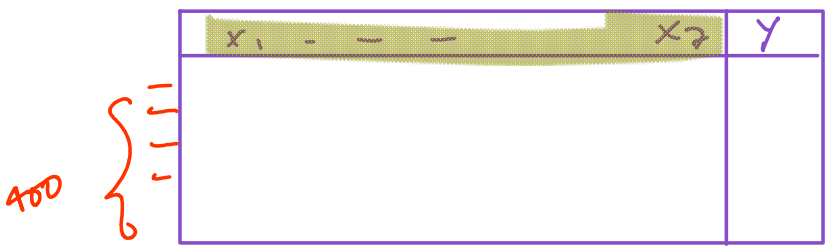
P.C.A →

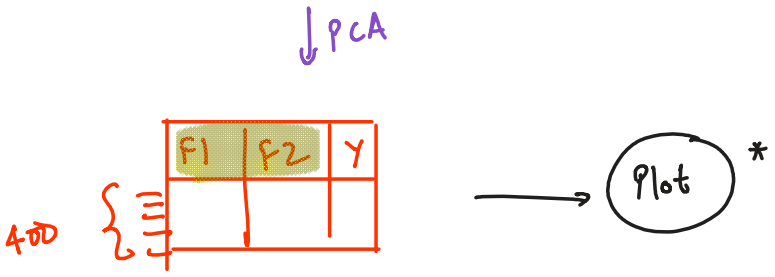
PC 1	PC 2

↓ Plot

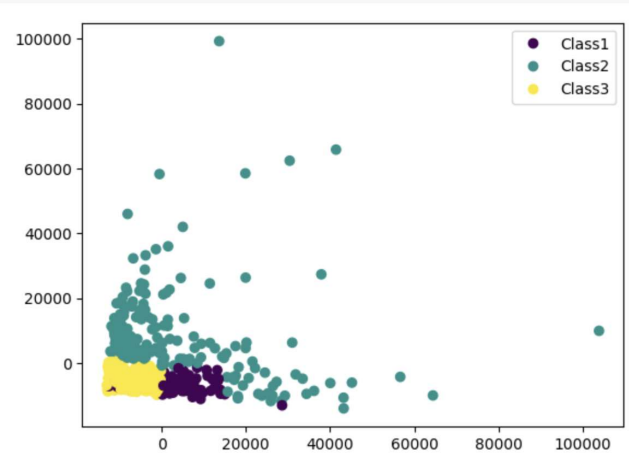








Q. Can normal log Reg work ?



OVR
Multi class
Non linear

How will Logistic Regression handle non-linear,multi-class data ?

4 options

Active Duration (Most preferred: 30 seconds)

Appears for 45 Secs

Non Linear
multi class

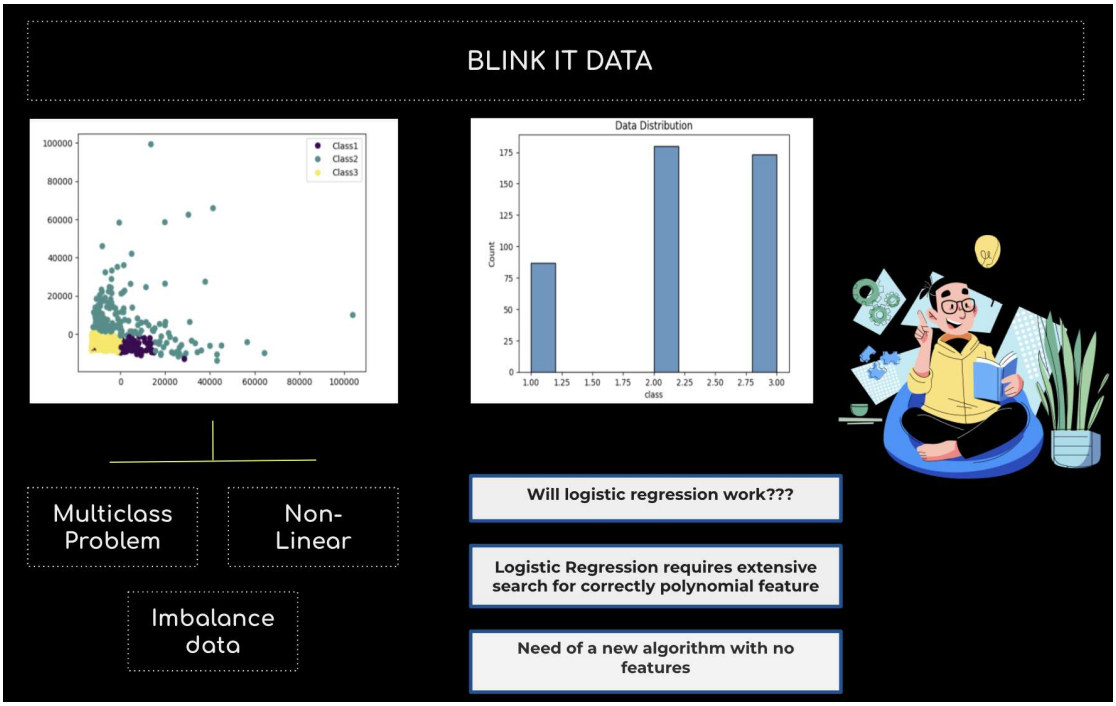
A Polynomial, OneVsRest

B Linear, OneVsRest

C OneVsRest, Polynomial

D OneVsRest, Linear

Polynomial features
(multiple data points)
(20 features
Polynom degree = 3

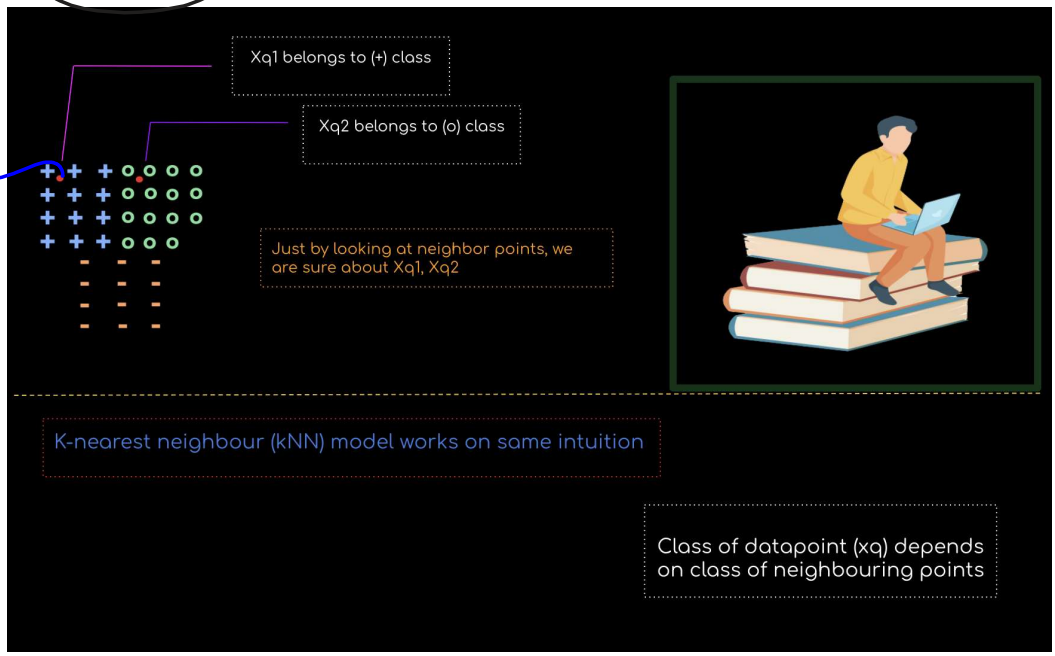


So there is need for some better Algorithms

→ (K - Nearest Neighbours)

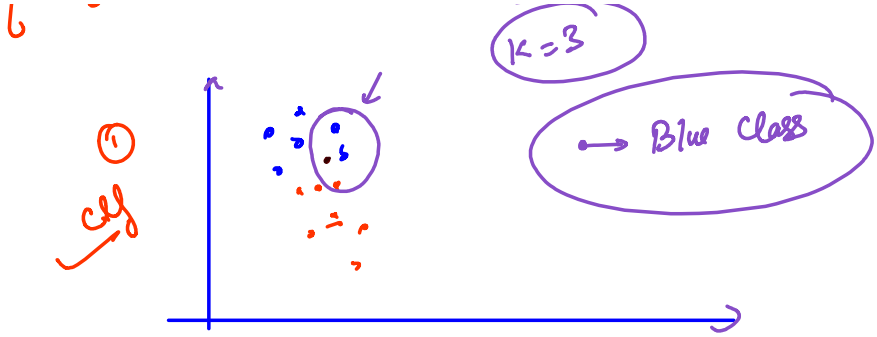
(KNN → A point is similar to its neighbours)

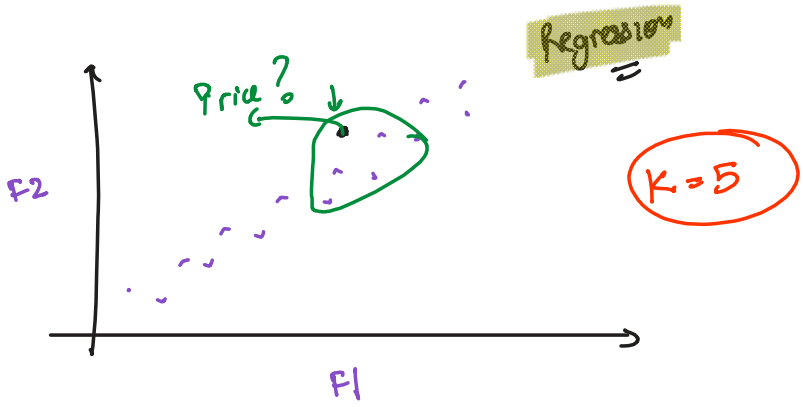
K = 3 *



→ [Regression + Classification]

K = 3



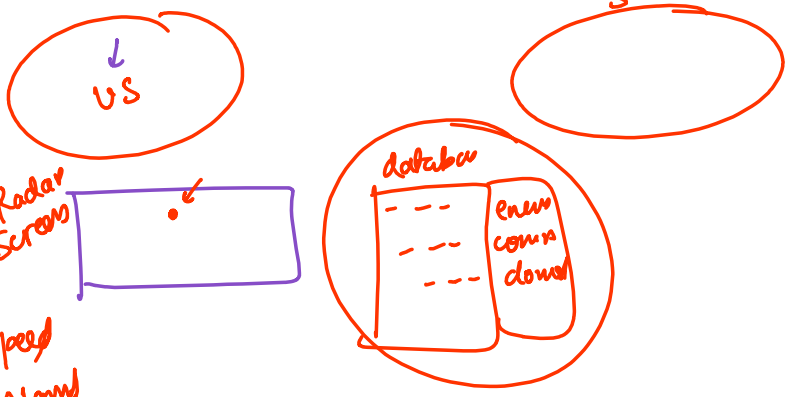


KNN → Nut and Bolt

History → KNN 1950(s) + [US & Soviet Union]

- enemy plane
- commercial plane
- domestic plane

(Radar)



- speed
- Algmd
- direction
- Accelerate

{Why are we still learning this}?

→ [KNN → is used in a lot of intermediate tasks]

* [KNN → is used in a lot of intermediate tasks]

A Step by Step Example

①

	x_1	x_2	Y	$d^*_{x_q}$
R1	2	5	C	4
R2	0	1	E	2
R3	3	4	E	1
R4	5	2	C	5

Steps ↓

x_q • [2, 1]

① Dist (x_q with 1, 2, 3, 4, 5)

② Sort distance in ascending order

Euclidean dist x_q and R_1 R_2 R_3 R_4 R_5

$x_q \rightarrow R_1 = \sqrt{(2-2)^2 + (5-1)^2} = 4$

sort (4, 2, 1, 5) → → R_3 R_2 R_1 R_4
[1, 2, 4, 5]

K = 2 *

x_q is similar to R_2 and R_3
↓ ↓
E E

= Enemy

How does kNN work?

If xq = [2,5] & data contains 6 data points:

Step 1 : Find euclidean distance:

(Training)

	f1	f2	y
X1	3	6	1
X2	6	4	1
X3	8	2	3
X4	7	5	3
X5	1	4	2
X6	2	2	2

	f1	f2	?
X1	3	6	1.41
X2	6	4	3.00
X3	8	2	6.48
X4	7	5	5.00
X5	1	4	1.41
X6	2	2	2.00



1. **Step1:** We find the **Euclidean distance between** $x_q = [2, 5]$ **and all the training data.**

	f1	f2	y	$d^{(i)} = \sqrt{(x^q_{f1} - x^{(i)}_{f1})^2 + (x^q_{f2} - x^{(i)}_{f2})^2}$
$x^{(1)}$	3	6	1	$\sqrt{(1^2 + 1^2)} = 1.41$
$x^{(2)}$	6	4	1	$\sqrt{(4^2 + 1^2)} = 3.00$
$x^{(3)}$	8	2	3	$\sqrt{(6^2 + 3^2)} = 6.48$
$x^{(4)}$	7	5	3	$\sqrt{(5^2 + 0^2)} = 5.00$
$x^{(5)}$	1	4	2	$\sqrt{(1^2 + 1^2)} = 1.41$
$x^{(6)}$	2	2	2	$\sqrt{(0^2 + 2^2)} = 2.00$

x_q → unknown (query)

Step 2 : Sort data based on distance:

	f^1	f^2	y
X^1	3	6	1.41
X^2	6	4	3.00
X^3	8	2	6.48
X^4	7	5	5.00
X^5	1	4	1.41
X^6	2	2	2.00

Sorted →

	f^1	f^2	y	Distance
X^1	3	6	1	1.41
X^5	1	4	2	1.41
X^6	2	2	2	2.00
X^2	6	4	1	3.00
X^4	7	5	3	5.00
X^3	8	2	3	6.48

K=3

2 → Class 2

1 → Class 1



Assigned Label X_q → Class 2

Step 3 : Pick 3 data points having minimum distance:

	f^1	f^2	y	y
X^1	3	6	1	1.41
X^5	1	4	2	1.41
X^6	2	2	2	2.00
X^2	6	4	1	3.00
X^4	7	5	3	5.00
X^3	8	2	3	6.48

→ minimum distance from xq



Step 4 : Find majority class of these selected data points —>> class label for xq

	f^1	f^2	y	y
X^1	3	6	1	1.41
X^5	1	4	2	1.41
X^6	2	2	2	2.00

→ Majority class 2

↓

Xq belongs to class 2



This selection of data points is decided by Hyperparameter "k", hence the name kNN

Quiz → Algo Pseudo Code

Break → 10:11 pm *

Arrange the statements in correct order based on kNN algo

s1- find majority vote

s2- perform euclidean distance

s3- sort and select k datapoints

s4- give class to x_q datapoint

[s2, s3, s1, s4]

4 options

Active Duration(Most preferred: 30 seconds)

Appears for | 60 Secs

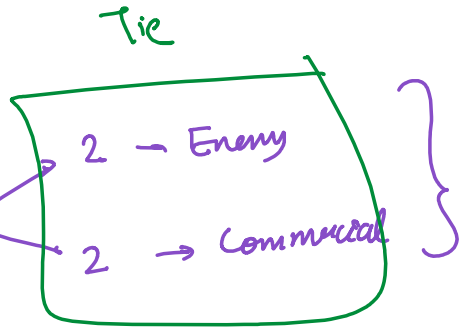
A | s2,s1,s3,s4

B | s2,s3,s4,s3

C | s2,s3,s1,s4

D | s2,s4,s3,s1

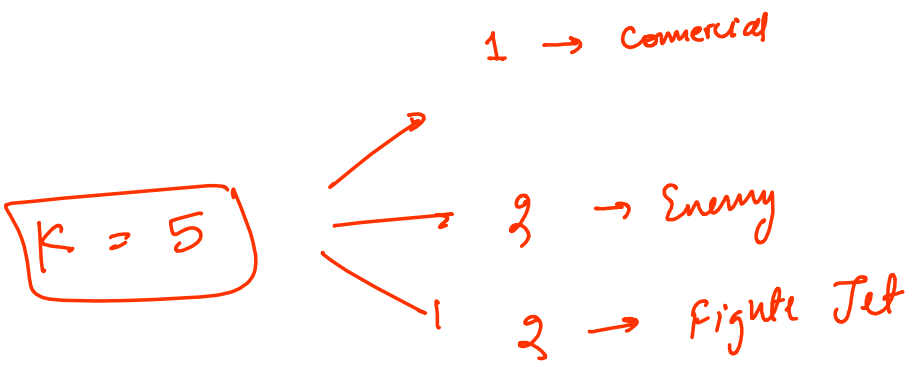
[small talk on K value]



* (K = even)

K = odd (✓) preferred

Qus Can there be a tie in K = odd ?



If x1 at (4,0) , x2 at (0,1) and x3 (5,0) and xq at (0,0). Which is nearest point to xq ?

x_q = Xval.[30] = (-0.7, 1, 2, 0, 0, 0, 2.5)

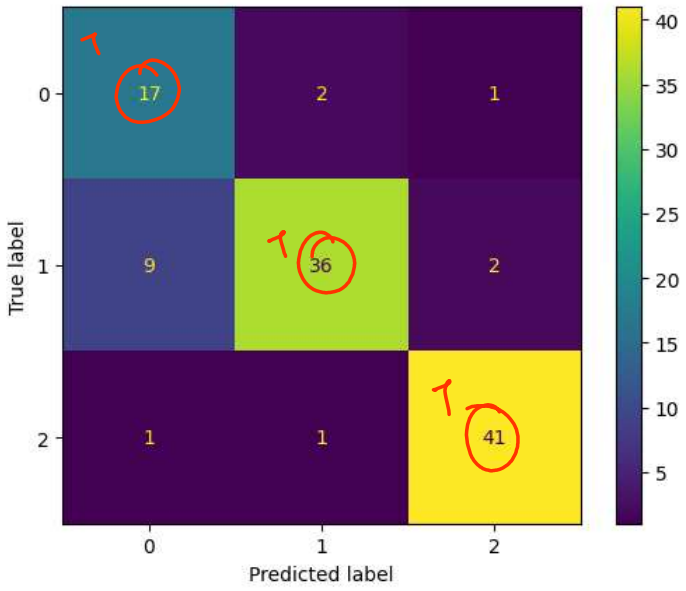
x_3 = (0.2, 0.1, 0, 0, 0, 5, 7)

$\sqrt{(0.2+0.7)^2 + (1-0.1)^2 + (2-0)^2 + \cdots}$

$= \sqrt{5}$

$\sqrt{(6.2+0.7)} + (1-0.1)^2 (2.3)$

$= 5$

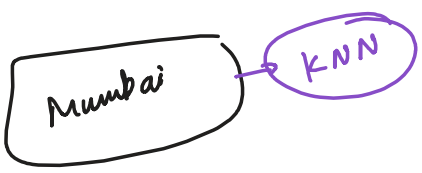


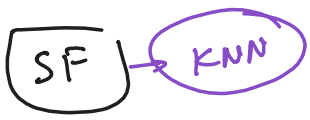
* [Assumptions of KNN]
(70 - 80%)

Is your data following the principle → Close data pts are similar

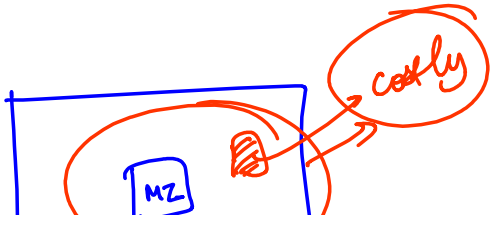
As kNN works on the principle that:

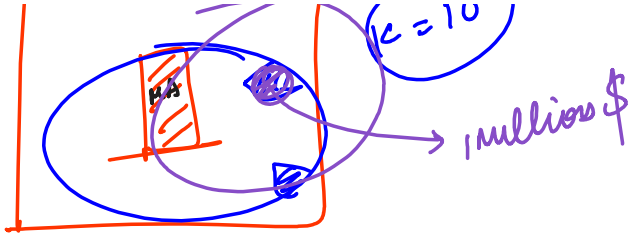
- All the **characteristic the majority nearest neighbors have, same** should be for the **query datapoint**
- hence **kNN assumes that the neighborhood** for a datapoint must be **homogeneous in nature (having same properties)**

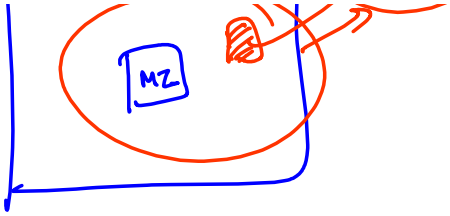









K = 10
millions $


MZ